

Подстановка $x = 2 \cos \theta$ и многочлены Чебышёва

Домашнее задание

1 сентября 2026 года

Теоретический минимум. $T_n(\cos \theta) = \cos n\theta$, $U_{n-1}(\cos \theta) = \sin n\theta / \sin \theta$, the recurrence $T_{n+1} = 2xT_n - T_{n-1}$ and the hyperbolic branch $T_n(\cosh t) = \cosh nt$; Chebyshev's extremal theorem; the inequalities of Markov and Bernstein; orthogonality with weight $(1-x^2)^{-1/2}$; the semicircle substitution $x = 2 \cos \theta$ in three-term recurrences.

1. Prove that $T_m \circ T_n = T_{mn}$ and that these are, up to affine conjugacy, the only polynomial families of degree > 1 closed under composition together with the powers x^n (теорема Ритта, достаточно сформулировать и проверить оба примера).
2. Prove the orthogonality relations $\int_{-1}^1 \frac{T_m T_n}{\sqrt{1-x^2}} dx = 0$ for $m \neq n$, and expand x^n in the basis $T_0, \dots, T_n$.
3. Prove that the roots of U_{n-1} are $\cos \frac{k\pi}{n}$ and deduce that the eigenvalues of the tridiagonal matrix with 0 on the diagonal and 1 on the neighbouring diagonals are $2 \cos \frac{k\pi}{n+1}$; compare with the discrete Wirtinger inequality from the first sheet.
4. Prove that $\min_p \max_{[a,b]} |p| = 2\left(\frac{b-a}{4}\right)^n$ over monic p of degree n , and find the polynomial attaining it.
5. (Proposed problems, №235) Prove that $\cos \frac{\pi}{7}$ has minimal polynomial $x^3 - \frac{1}{2}x^2 - \frac{1}{2}x + \frac{1}{8}$ over $\mathbb{Q}$, that $\cos \frac{2\pi}{7}$ has minimal polynomial $x^3 + \frac{1}{2}x^2 - \frac{1}{2}x - \frac{1}{8}$, and that $\cos \frac{3\pi}{7}$ has the same minimal polynomial as $\cos \frac{\pi}{7}$.
6. Prove that $2 \cos \frac{2\pi}{n}$ is a rational number only for $n \in \{1, 2, 3, 4, 6\}$, using the recurrence $T_{n+1} = 2xT_n - T_{n-1}$ and the rational root theorem.

7. Let p be a real polynomial of degree n with $|p(x)| \leq 1$ on $[-1, 1]$. Prove that $|p(x)| \leq |T_n(x)|$ for every $|x| > 1$, and that the leading coefficient of p is at most 2^{n-1} in absolute value.
8. Prove that $\sum_{k=0}^n T_k(x)t^k = \frac{1-tx}{1-2tx+t^2} + O(t^{n+1})$, i.e. find the generating functions of T_n and U_n , and deduce the addition formula $T_{m+n} + T_{m-n} = 2T_m T_n$.
9. (Solve in radicals) Solve the equation $x^3 - 3x = c$ for every real c by the substitution $x = 2\cos\theta$ when $|c| \leq 2$ and $x = 2\cosh t$ otherwise; explain what the casus irreducibilis of Cardano's formula looks like in this language.
10. Prove the Chebyshev bound: for any n distinct points of $[-1, 1]$ the Lebesgue constant of interpolation is at least $c \ln n$, and that the Chebyshev nodes $\cos \frac{(2k-1)\pi}{2n}$ achieve the order $\ln n$.